Simple Linear Regression: Equations and Step-by-Step Calculation

1. Dataset

x	y
1	2
2	3
3	5

2. Linear Regression Model

We aim to fit the model:

$$\hat{y} = a + bx$$

where:

- a is the intercept
- b is the slope

3. Mean-Based (Centered) Formulas

- Mean of x and y:

$$\bar{x} = \frac{1}{n}\sum x_i, \quad \bar{y} = \frac{1}{n}\sum y_i$$

- Slope:

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

- Intercept:

$$a = \bar{y} - b\bar{x}$$

4. Derivation of Raw-Score Formulas

Slope Derivation

Start with:

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

Expand numerator:

$$\sum(x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y}$$

Expand denominator:

$$\sum(x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2$$

Substitute:

$$b = \frac{\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}$$

Intercept Derivation

$$a = \bar{y} - b\bar{x} = \frac{\sum y_i}{n} - b \cdot \frac{\sum x_i}{n}$$

5. Compute Means

$$\bar{x} = \frac{1+2+3}{3} = 2, \quad \bar{y} = \frac{2+3+5}{3} = \frac{10}{3} \approx 3.33$$

6. Centered Method Calculation

Slope b

$$\begin{aligned}\sum(x_i - \bar{x})(y_i - \bar{y}) &= (1-2)(2-3.33) + (2-2)(3-3.33) + (3-2)(5-3.33)\\ &= (-1)(-1.33) + (0)(-0.33) + (1)(1.67) = 1.33 + 0 + 1.67 = 3\\ \sum(x_i - \bar{x})^2 &= (1-2)^2 + (2-2)^2 + (3-2)^2 = 1 + 0 + 1 = 2\end{aligned}$$

$$b = \frac{3}{2} = 1.5$$

Intercept a

$$a = \bar{y} - b\bar{x} = 3.33 - 1.5 \cdot 2 = 3.33 - 3 = 0.33$$

7. Raw-Score Method Calculation

Step 1: Compute Sums

$$\sum x_i = 6, \quad \sum y_i = 10, \quad \sum x_i y_i = 23, \quad \sum x_i^2 = 14$$

Slope b

$$b = \frac{3 \cdot 23 - 6 \cdot 10}{3 \cdot 14 - 6^2} = \frac{69 - 60}{42 - 36} = \frac{9}{6} = 1.5$$

Intercept a

$$a = \frac{10}{3} - 1.5 \cdot 2 = 3.33 - 3 = 0.33$$

8. Final Regression Line

$$\hat{y} = 1.5x + 0.33$$

Prepared By:

Md. Atikuzzaman

Lecturer

Department of Computer Science and Engineering

Green University of Bangladesh

Email: atik@cse.green.edu.bd